

Phase 3: Project Design

1. System Architecture

The application follows a **Client-Server Architecture**:

Client Side (Frontend): Developed using **HTML, CSS, and Bootstrap**. It provides an interactive web interface where users can enter the required Human Development indicators and view prediction results.

Server Side (Backend): A **Python Flask** application (`app.py`) processes user requests, validates the input values, performs necessary preprocessing, and sends the data to the trained Machine Learning model.

Machine Learning Model: A trained and serialized Machine Learning model (`hdi_model.pkl`) is loaded into memory by the Flask application. The model predicts the **HDI Score** and classifies it into one of the four categories: **Very High, High, Medium, or Low Human Development**.

2. Directory Structure Strategy

To ensure smooth development and deployment, the project follows a well-organized directory structure. Core files such as **app.py**, **requirements.txt**, and the **models** folder remain in the root directory to support deployment on cloud platforms like **Render** or **Heroku**.

```
HDI_Prediction_System/
|
├── Dataset/
|   └── Human Development Index.csv
|
├── models/
|   └── hdi_model.pkl
|
├── static/
|   └── style.css
|
├── templates/
|   ├── home.html
|   └── result.html
|
├── app.py
├── requirements.txt
└── README.md
```



3. Database Schema (Conceptual ER Diagram)

Entity Name	Attributes	Description
Users	UserID (PK), Name, Email, Password	Stores user information for future authentication.
HDI_Data	DataID (PK), UserID (FK), LifeExpectancy, MeanYearsSchooling, ExpectedYearsSchooling, GNI_PerCapita	Stores the input values entered by users.
ML_Model	ModelID (PK), ModelName, Algorithm, Accuracy, ModelFile	Stores information about the trained Machine Learning model.
Prediction_Result	PredictionID (PK), DataID (FK), ModelID (FK), HDI_Score, HDI_Category, PredictionDate	Stores the predicted HDI score and development category.

4. UI/UX Design Flow

Home Page: Displays an introduction to the Human Development Index Prediction System with a button to start prediction.

Input Form: Provides a user-friendly form where users enter the following values:

- Life Expectancy
- Mean Years of Schooling
- Expected Years of Schooling
- GNI per Capita (PPP)

Prediction Page: The system processes the input using the trained Machine Learning model.

Result Page: Displays the predicted **HDI Score**, **HDI Category (Very High, High, Medium, or Low)**, and a brief explanation of the country's development status.

This follows the same format as your sample while being customized for your **Human Development Index (HDI) Prediction System**.